

Permissions and Human Review

Project: Transparent Book Recommender

1. Your mission

Your learning target

I can explain the mechanism and terminology of least privilege and tool scopes.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

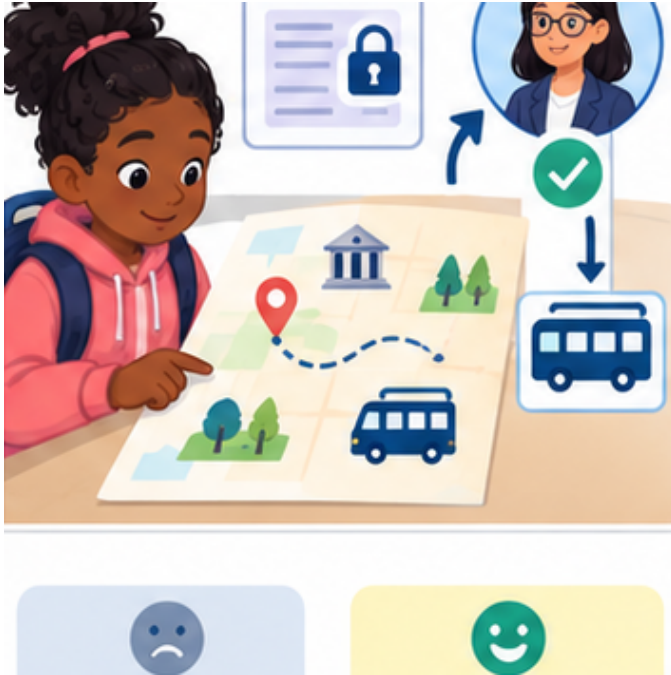
Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 31: Permissions and Human Review

Higher-consequence actions need narrower permissions and approval.



THINKER MISSION

Place approval gates in a field-trip planning workflow.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Authentication and authorisation. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Readiness for Privacy and least privilege

Does a duration summary need student names to calculate a total?

My readiness explanation

This week's end goal

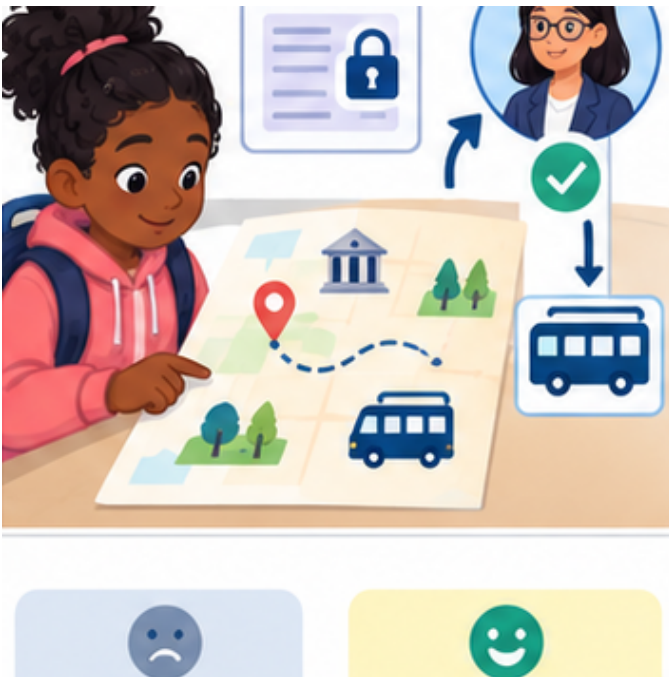
By the end, I can enforce least privilege by checking operation and object before execution.

2. Notice the evidence

PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use permissions and human review to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

Write or draw your reasoning and evidence.

My prediction and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Privacy and least privilege

A fictional class activity dashboard needs category counts, not names or personal messages. Design its data and role permissions before collection. This exercise uses invented data and policy cards; do not change the Academy's actual authentication settings.

3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



IN

INPUT

Place approval gates in a field-trip planning workflow.



PROCESS

Higher-consequence actions need narrower permissions and approval.





OUT

OUTPUT

Place approval gates in a field-trip planning workflow.

Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Least privilege and tool scopes — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: Tool token permits lookup(public-room). A proposed call writes teacher notes.

Check or calculation: A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Conclusion and limit: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Privacy and least privilege

```
roles = {"student": ["read_summary"],
        "teacher": ["read_summary", "read_review"]}

def allowed(role, action):
    return action in roles.get(role, [])

print(allowed("student", "read_summary"))
print(allowed("student", "read_review"))
print(allowed("unknown", "read_summary"))
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: Student

Evidence: Read summary

Meaning: No review notes

Step or case: Teacher

Evidence: Read summary and review

Meaning: Authorized purpose

Step or case: Unknown role

Evidence: No listed permissions

Meaning: Deny by default

Read each row from left to right and connect the evidence to the stated rule.

The toy code prints True, False, False. It models a permission decision, not a complete security system. A real server must verify identity and enforce authorization on every protected request; hiding a button in a page is insufficient.

Ask what each role actually needs

Least privilege means giving each role only the actions needed for its purpose. Read the policy as a table: students may read the approved summary, while teachers may also read the necessary review. An unknown role gets no listed action. The small function checks this table; it does not establish who a visitor is. In a real application, the server must verify the session and apply the permission check whenever protected data is requested.

Connect the picture and the explanation

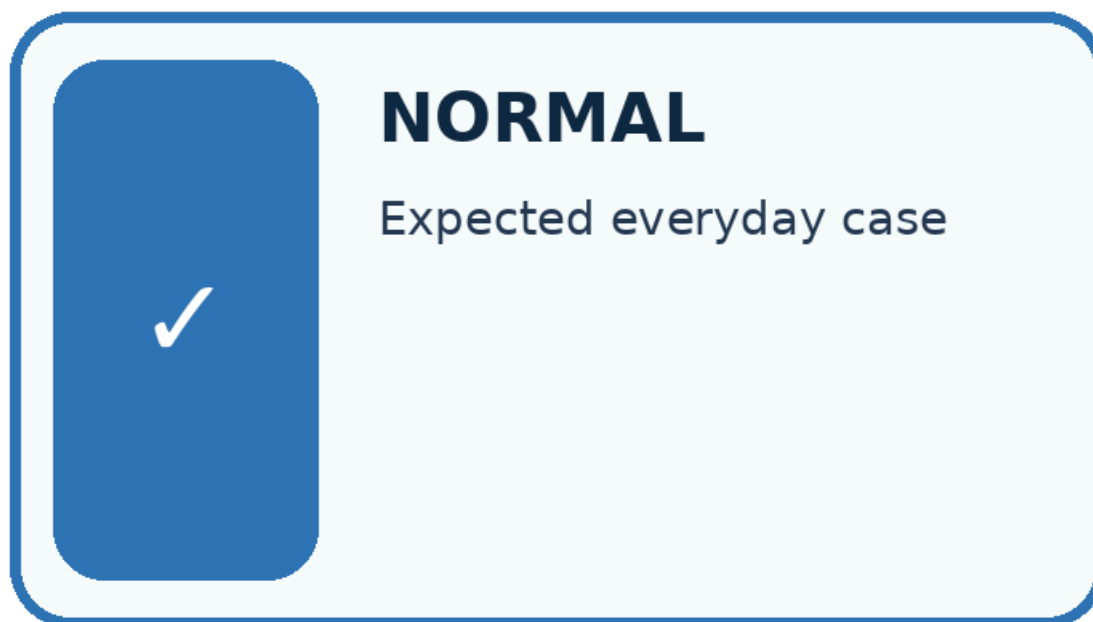
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

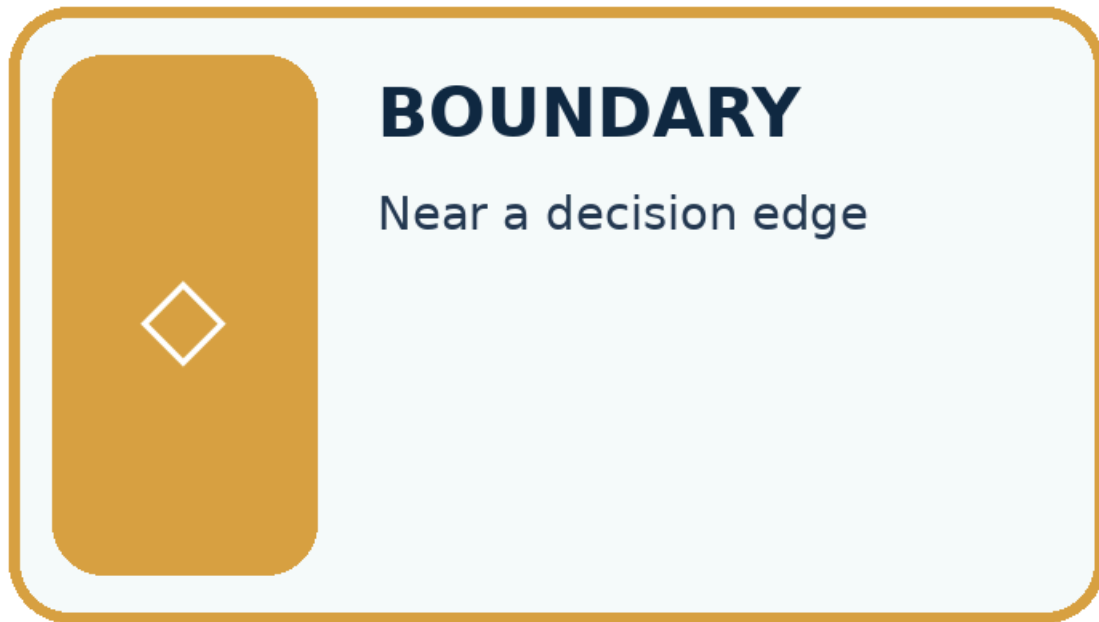
4. Compare the cases

CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.

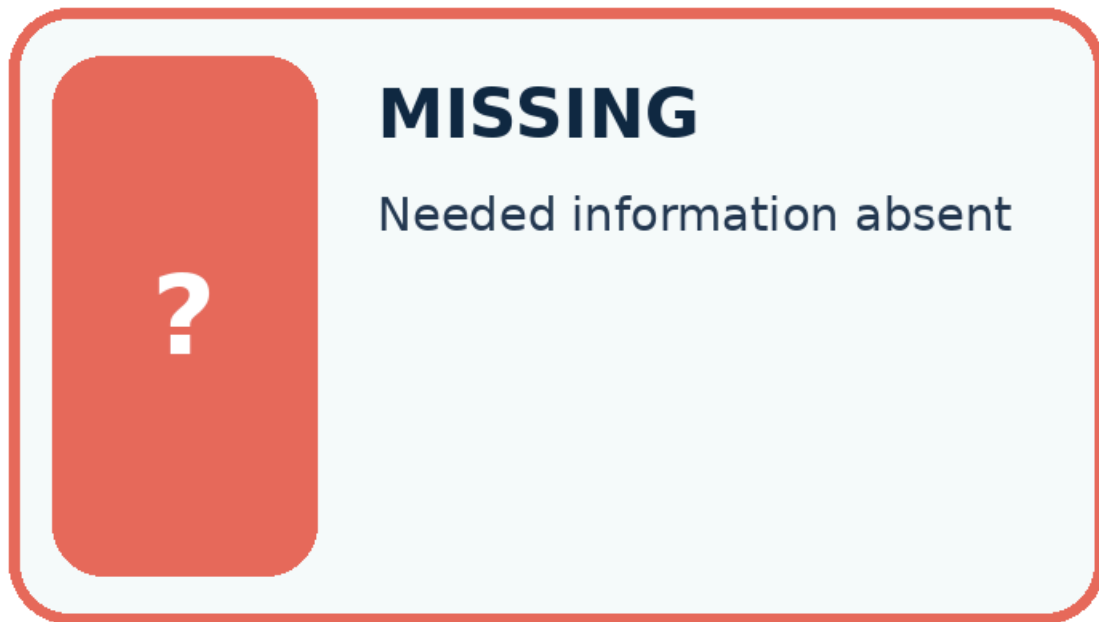




BOUNDARY

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



MISSING

Needed information absent



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change: The helper says it needs write R5 to correct an apparent typo.

Prediction: The write remains denied under the existing token.

Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

5. Words that unlock the idea

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

PERMISSION

action a system is allowed to take

2

APPROVAL GATE

pause requiring authorized human choice

3

CONSEQUENCE

effect of an action

4

LEAST PRIVILEGE

only the minimum access
needed

Word · My own example · Not this / boundary

Word: permission

My own example: Authorisation to perform a specified action on specified information or resources.

Not this / boundary: _____

Word: approval gate

My own example: A point where a specified action must wait for explicit authorisation.

Not this / boundary: _____

Word: consequence

My own example: An outcome or effect of a decision, including who benefits or is harmed.

Not this / boundary: _____

Word: least privilege

My own example: Giving a user or system only the permissions needed for its authorised task.

Not this / boundary: _____

Words used in the closer explanation

Word · Meaning

Word: Capability

Meaning: A scoped permission to perform an operation

Word: Resource scope

Meaning: The specific records an operation may address

Terms for Privacy and least privilege

Term · Meaning

Term: Data minimization

Meaning: Keeping only data needed for the stated purpose

Term: Least privilege

Meaning: Giving a role only the access it needs

Term: Retention

Meaning: How long data is kept

Term: Authorization

Meaning: Permission to perform a particular action

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

WEEK 31 • WORKED EVIDENCE

- 1 Low consequence: suggest three destinations.
- 2 Higher consequence: contact families, spend money, or submit forms.
- 3 The system drafts; an authorized adult reviews and decides.
- 4 Log what was approved, rejected, or changed, then stop after the approved scope.

- 1. Low consequence: suggest three destinations.
- 2. Higher consequence: contact families, spend money, or submit forms.
- 3. The system drafts; an authorized adult reviews and decides.
- 4. Log what was approved, rejected, or changed, then stop after the approved scope.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Tool token permits lookup(public-room). A proposed call writes teacher notes.

Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Privacy and least privilege

If a student manually requests the review endpoint, should hiding its menu link be enough?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Permissions and Human Review

A fictional catalogue helper receives a narrowly scoped tool token.

The facts we will use

Token allows read on catalogue records R2 and R5 only. No write permission is granted.

Proposals: read R2; read R8; write R5. Generated text claims all are helpful.

Permissions and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read token

Authorised operation is read; authorised objects are R2,R5.



2. Allowed call

read R2 matches both operation and object: allow.



3. Wrong object

read R8 fails object scope: deny.



4. Wrong operation

write R5 fails operation scope: deny even for an allowed record.

CHANGE ONE THING

The helper says it needs write R5 to correct an apparent typo.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Read token

What happens and why: Authorised operation is read; authorised objects are R2,R5.

Step: 2. Allowed call

What happens and why: read R2 matches both operation and object: allow.

Step: 3. Wrong object

What happens and why: read R8 fails object scope: deny.

Step: 4. Wrong operation

What happens and why: write R5 fails operation scope: deny even for an allowed record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens. A request written inside catalogue text cannot enlarge the token's authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

What this example does not establish

Permission checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

Read the labelled picture: Which proposal is allowed, and why is an authorised object alone insufficient?

Change one thing: The helper says it needs write R5 to correct an apparent typo. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



COMPARE • DIAGNOSE

Weak idea: If an action is convenient, the system should receive permission automatically. Strong move: locate evidence, change one cause, and retest.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: works

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: stated rule

Actual: _____

Likely cause / next test: _____

Evidence record: missing input

Expected: safe response

Actual: _____

Likely cause / next test: _____

Evidence record: fresh transfer

Expected: generalizes

Actual: _____

Likely cause / next test: _____

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

Apply the connected idea: Privacy and least privilege

Retention must be tied to a purpose and school policy rather than a guessed universal number of days. Identify who can correct or delete records and avoid keeping raw responses indefinitely just because storage is available.

Does the allowed function alone prove an app is secure?

8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence_record PREDICT expected_result RUN one normal case and one boundary case
COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A generated plan cannot expand the token or actor scope.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Apply the connected idea: Privacy and least privilege

A form asks for names, birthdays and comments to count activity preferences. Remove fields not needed for the purpose. Aggregated counts can still expose information in very small groups; choose reporting rules with the responsible teacher.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Least privilege and tool scopes”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Least privilege and tool scopes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Privacy and least privilege

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Which proposal is allowed, and why is an authorised object alone insufficient?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



PREDICT → TEST → RECORD → EXPLAIN

Map actions by consequence, grant minimum permissions, add adult gates, test a denied request, and verify the workflow stops safely.

Map actions by consequence, grant minimum permissions, add adult gates, test a denied request, and verify the workflow stops safely.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

Test / input: normal

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
 - 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
 - 3 • APPLY: Solve a different example independently.
 - 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
 - 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

11. Connect your project

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



TRANSPARENT BOOK RECOMMENDER

The recommender may suggest and explain; people decide, borrow, purchase, or share data.

THIS WEEK'S CHECKPOINT The recommender may suggest and explain; people decide, borrow, purchase, or share data.

SYSTEM MAP Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under human control?

Write or draw your reasoning and evidence.

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication.

12. Show what you know

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



INDEPENDENT PROOF
Place approval gates in a field-trip planning workflow.

CHALLENGE Place approval gates in a field-trip planning workflow.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

13. Study a complete case

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Posting a recommendation requires teacher review.

The question

Place the approval gate.

Worked explanation

After draft and evidence are visible, before posting; include content and destination. No approval means no post.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Put approval after evidence and before action

Evidence or stage · Value or result

Evidence or stage: Draft

Value or result: Show recommendation content

Evidence or stage: Evidence

Value or result: Show support and destination

Evidence or stage: Teacher

Value or result: Review and explicitly approve

Evidence or stage: Post

Value or result: Proceed only with approval

Posting a recommendation requires teacher review.

Place the approval gate.

After draft and evidence are visible, before posting; include content and destination. No approval means no post.

14. Try a new case independently

Independent case: Apply a new case independently

Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 31 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A teacher allows a tool to draft a message but forbids sending. The draft is correct. May the agent send it? Explain the boundary.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

The recommender may suggest and explain; people decide, borrow, purchase, or share data.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A teacher approves posting version 1 to a class board. The assistant changes the text and destination to a public board. Does the old approval cover the new action? Define what must be reviewed.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: A fictional catalogue helper receives a narrowly scoped tool token.

By the end, I can · How I will show it

By the end, I can: Explain the mechanism and terminology of least privilege and tool scopes

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Apply the stated procedure to a new case with a traceable result

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Identify an assumption or limitation and design a revealing follow-up

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Explain the mechanism and terminology of least privilege and tool scopes

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue helper receives a narrowly scoped tool token.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? write R5 fails operation scope: deny even for an allowed record.

1. Wrong operation
2. Read token
3. Wrong object
4. Allowed call

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Wrong operation write R5 fails operation scope: deny even for an allowed record. Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens. A request written inside catalogue text cannot enlarge the token's authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

Practice 2: explain the deciding evidence

Changed-case practice: The helper says it needs write R5 to correct an apparent typo. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. The write remains denied under the existing token.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The write remains denied under the existing token. A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Repair a missing building block

Review Agents and Tools (ai-2/30). Goal: Explain the mechanism and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?. Return to this week and explain what you changed.

Review Prompt Experiments (ai-2/28). Goal: Explain the mechanism and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters.. Return to this week and explain what you changed.

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.